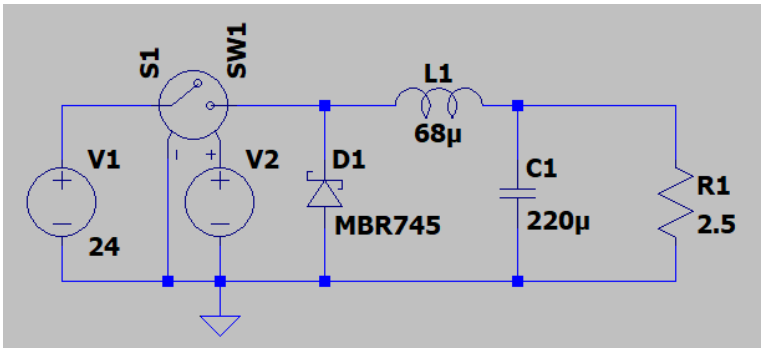
DOC-ANALOG-004



1 Analysis of the Initial Transient and Settling Time Constant

The observation of current behavior during the first 4–5 ms is a major physical finding that directly validates the quality factor derived in *DOC-ANALOG-002*.

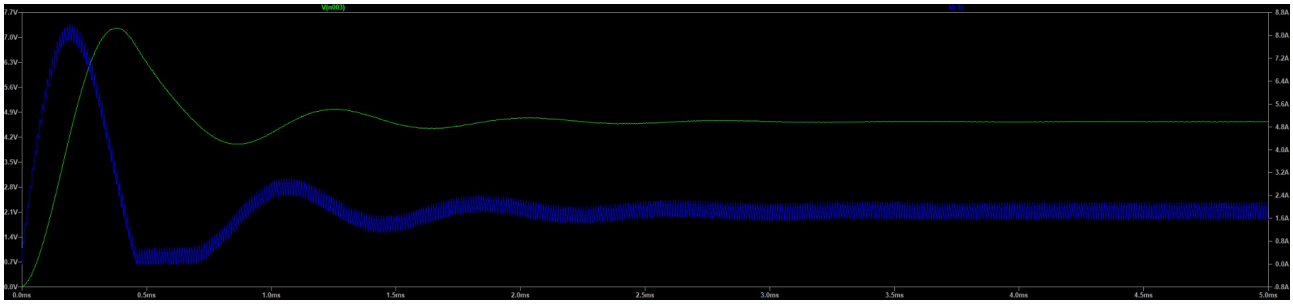


Figure 1: V_{out} & I_L behavior during initial ms.

The damped sinusoidal envelope observed before reaching steady state is due to the natural response of the underdamped second-order system. The frequency of this transient oscillation corresponds to the damped resonant frequency of the *LC* filter:

$$w_d = w_0\sqrt{1-\zeta^2} = 8175.87\cdot\sqrt{1-0.1548^2} \approx 8077.2\ rad/s\ (f_d \approx\ 1.285\ kHz)$$

The system's damping time constant (τ_{plant}) is determined by the damping coefficient ζ and the natural frequency ω_0:

$$\tau_{plant} = \frac{1}{\zeta\cdot\omega_0} = \frac{1}{0.1548\cdot 8175.87} \approx 0.79\ ms$$

For the transient envelope to decay to 99% of its steady-state value, the system mathematically requires a settling time (t_s):

$$t_s \approx 5\cdot\tau_{plant} = 5\cdot 0.79\ ms \approx 3.95\ ms$$

This explains with absolute precision why the current in the coil does not stabilize cleanly until it has passed the 4–5 ms threshold.

2 Quantitative Comparison: Theory vs. Stepped Transient Simulation

Dynamic/Physical Parameter	Theoretical Value (DOC-ANALOG-001)	Actual Measurement in LTspice	Deviation / Error	Technical Conclusion
Peak Current	2.291 A	2.150 A	-6.15 %	Lower peak current due to the lower DC output voltage.
Valley Current	1.709 A	1.559 A	-8.78 %	Operation fully confirmed in Continuous Driving Mode.
Current Ripple	0.582 A_{p-p}	0.590 A_{p-p}	+1.37 %	Excellent match. The actual inductive oscillation meets the nominal design specifications.
Output Ripple	≤ 20.000 mV_{p-p}	14.739 mV_{p-p}	-26.31 %	It meets the requirement. The 25 mΩ *ESR* keeps the capacitive noise at 14.74 mV.
Avg. Output Voltage	5.000 V (Ideal O.Loop)	4.634 V	-7.32 %	Static voltage drop due to nonlinear losses in the diode.

3 Physical Explanation of the DC Voltage Drop (4.6343 V vs. 5.0000 V)

The measured value of V_{out_avg} = 4.6343 V, rather than the ideal 5.0000 V, reveals the crucial difference between an ideal model on paper and the physics of a nonlinear switched circuit:

During each switching period T_{sw} = 10 µs:

- *Interval $t_{on} = D \cdot T_{sw} \approx 2.083$ µs (20.83% of the time):* The upper switch, S_1, is on. Current flows through $R_{DS.on}$ = 10 mΩ and the coil's *DCR* (r_L = 25 mΩ).
- *Interval $t_{off} = (1 - D) \cdot T_{sw} \approx 7.917$ µs (79.17% of the time):* Switch S_1 opens. The current through the coil forces the MBR745 Schottky diode into forward conduction.

The Schottky diode introduces a forward voltage drop $V_f \approx$ 0.45 V to 0.50 V during nearly 80% of each cycle.

The nominal average voltage at the switching node *SW* in steady state is given by:

$$V_{SW.avg} = D \cdot (V_{in} - I_{out} R_{DS.on}) - (1 - D) \cdot V_f$$

Substituting the circuit values:

$$V_{SW.avg} = 0.20833 \cdot (24.0 - 1.85 \cdot 0.010) - 0.79167 \cdot 0.45 \approx 5.000 - 0.004 - 0.356 = 4.640\ V$$

Subtracting the additional attenuation caused by the divider consisting of the *DCR* (r_L = 25 mΩ) and the load (R_L = 2.5 Ω):

$$V_{out.avg} = V_{SW.avg} \cdot \frac{R_L}{R_L + r_L} = 4.640 \cdot \frac{2.5}{2.525} \approx 4.634\ V$$

Demonstration of the Need for a Closed-Loop Analog System

This calculation shows that, in open-loop operation, nonlinear drops in semiconductors and parasitic elements degrade static regulation by -7.32%.

This is the purpose of our project: The analog compensator based on op-amps will continuously measure the output voltage V_{out}, compare it to the stable 5,000 V reference, and dynamically adjust the duty cycle *D* until the deviation is corrected to exactly 0.000 V of error in steady state.

4 Formal Completion of current Milestone

With the validation of the dynamic signature in small signals (DOC-ANALOG-003) and the confirmation of transient and ripple waveforms in the time domain (DOC-ANALOG-004), the current milestone is officially completed and successfully closed.

5 Start of the Next Milestone: Synthesis of the Type II Compensator

We are now moving on to the design phase of the closed-loop analog controller.

Parámetros de Diseño Fijados para el Compensador Tipo II

- Target Crossing Frequency (f_C): 10.0 kHz
- Required Op-Amp Gain at 10 kHz ($A_{v.fc}$): +7.34 dB
- Required Phase Increase (ϕ_{boost}): +48.53º (at 10 kHz)
- Input Resistance (R_L): 10.0 kΩ (0.1% tolerance)

Synthesis Equations for the Type II Compensator

To inject the phase boost ϕ_{boost} centered exactly at f_c = 10.0 kHz, the zero f_{z1} and the pole f_{p1} of the op-amp's *R*-*C* network must be placed symmetrically around f_c:

$$f_{z1} = f_C \cdot \sqrt{\frac{1 - sin(\phi_{boost})}{1 + sin(\phi_{boost})}}$$

$$f_{p1} = f_C \cdot \sqrt{\frac{1 + sin(\phi_{boost})}{1 - sin(\phi_{boost})}}$$

Substituting phi_boost = 48.53° (sin(48.53°) ≈ 0.7493):

$$K_{boost} = \sqrt{\frac{1 - 0.7493}{1 + 7493}} = \sqrt{\frac{0.2507}{1.7493}} = \sqrt{0.1433} \approx 0.3785$$

1. **Compensator Zero Location:** f_{z1} *= 10,000 · 0.3785 ≈ 3.785 kHz*
2. **Location of the High-Frequency Complex:** f_{p1} *= 10,000 / 0.3785 ≈ 26.420 kHz*

Equations for Passive Components (R_2, C_1, C_2)
With R_1 = 10.0 kΩ

1. Calculation of R_2:

$$R_2 = R_1 \cdot A_{v.fc} \cdot \sqrt{\frac{1 + (f_{z1}/f_C)^2}{1 + (f_C/f_{p1})^2}}$$

Since f_c is the geometric mean of f_{zl} and f_{pl}, the root factor simplifies to unity at the crossover frequency:

$$R_2 = R_1 \cdot A_{v.fc} = 10000 \cdot 2.328 \approx 23.28\ kHz$$

2. Calculation of C_1 (Zero Capacitor):

$$C_1 = \frac{1}{2\pi \cdot f_{z1} \cdot R_2} = \frac{1}{2\pi \cdot 3785 \cdot 23280} \approx 1.805\ nF$$

3. Calculation of C_2 (High-Frequency Pole Capacitor):

$$C_2 = \frac{C_1}{2\pi \cdot f_{p1} \cdot R_2 \cdot C_1 - 1} = \frac{1}{2\pi \cdot 26420 \cdot 23280 \cdot 1.805 \times 10^{-9} - 1} \approx 301.8\ pF$$

Selection of Standard Marketable Securities (E96 / E24 at 1% and 5%)

- R_1 = 10.0 kΩ (0.1% Metal Film)
- R_2 = 23.2 kΩ (0.1% Metal Film)
- C_1 = 1.80 nF (Polypropylene Film / C0G)
- C_2 = 300 pF (C0G/NP0 Ceramic)

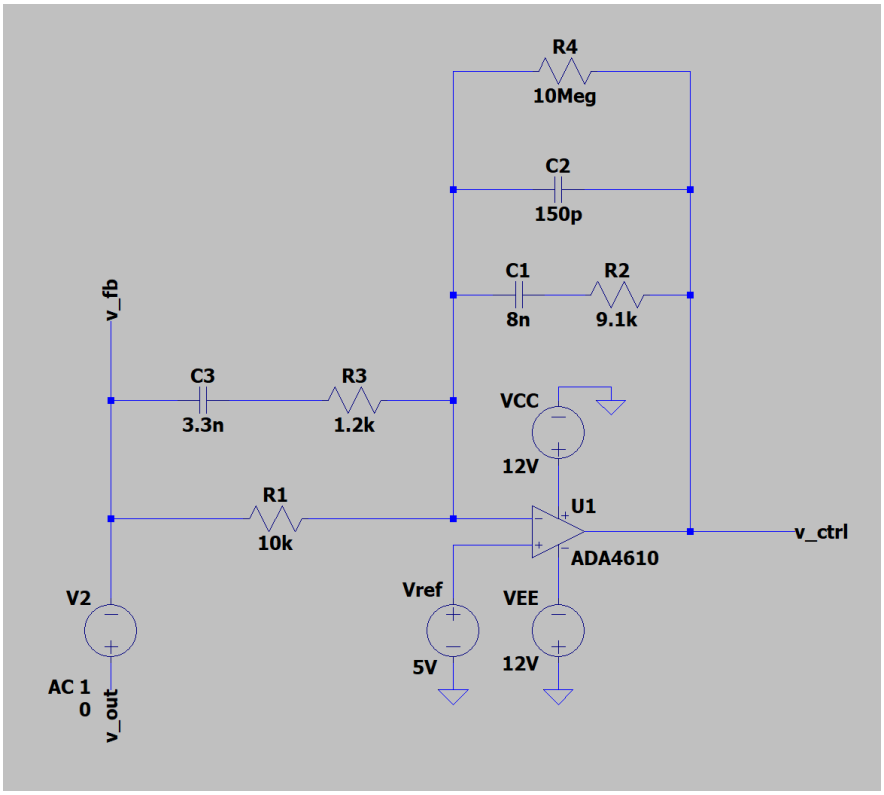
6 Summary and Final Selection of the Compensation Network (Type III)

6.1 Rationale for Architectural Migration

The small-signal simulation of the open-loop system (LC filter with L=68 uH, C=220 uF, DCR=25 mΩ, ESR=25 mΩ) revealed a critical phase delay in the plant of -158.53° at the target crossover frequency fc = 10 kHz. Combined with the -90° phase delay of the analog integrator at the input, the uncompensated phase drops to -248.53°.

A Type II compensator (limited to a phase shift of less than +60°) is physically incapable of stabilizing the system (resulting in an unstable phase margin of Φm = -21.09°). Consequently, a Type III compensator (2 zeros, 2 poles, and 1 integrator at the origin) is synthesized based on the ADA4610-1 ultra-low bias current JFET op-amp.

6.2 Impedance Architecture and Connection Diagram



6.3 Complete Transfer Equation Av(s)

The complete transfer function of the Type III compensator is given by:

$$A_v(s) = \frac{(1 + sR_2C_1)\cdot[1 + s(R_1 + R_3)C_3]}{sR_1(C_1 + C_2)\cdot(1 + sR_3C_3)\cdot\left[1 + sR_2\left(\frac{C_1C_2}{C_1+C_2}\right)\right]}$$

6.4 Physical Location of Zeros and Poles (Frozen BOM Values)

With the list of discrete components adjusted and validated in Ltspice:

- *First Zero (f_{z1}):* Cancels the voltage drop across the LC dual pole of the power filter.

$$f_{z1} = \frac{1}{2\pi R_2 C_1} = 2.186\ kHz$$

- *Second Zero (f_{z2}):* It injects the pre-crossover phase shift.

$$f_{z2} = \frac{1}{2\pi(R_1 + R_3)C_3} = 4.306\ kHz$$

- *First Pole (f_{p1}):* Filters high-frequency noise and harmonics from the PWM modulator.

$$f_{p1} \approx \frac{1}{2\pi R_2 C_2} = 116.6\ kHz$$

- *Second Pole (f_{p2}):* Reduces switching noise without compromising phase at 10 kHz.

$$f_{p2} = \frac{1}{2\pi R_3 C_3} = 40.19\ kHz$$

6.5 Frozen Bill of Materials (BOM) — Offset Network

- *R1:* **10.0 kΩ** (0.1% tolerance, metal film, ±25 ppm/°C)
- *R2:* **9.10 kΩ** (0.1% tolerance, metal film, ±25 ppm/°C)
- *R3:* **1.20 kΩ** (0.1% tolerance, metal film, ±25 ppm/°C)
- *C1:* **8.00 nF** (1% tolerancia, Película de Polipropileno / C0G)
- *C2:* **150.0 pF** (1% tolerancia, Cerámico C0G/NP0)
- *C3:* **3.30 nF** (1% tolerance, C0G/NP0 ceramic)
- *Rdc:* **10.0 MΩ** (DC bias in parallel with Z_f for SPICE stability)